

PAUL WANG, MECHANICAL ENGINEERING STUDENT

<https://paulhanwang.co.uk> • paulhanwang@gmail.com • [linkedin.com/in/paulhanwang/](https://www.linkedin.com/in/paulhanwang/)

WORK EXPERIENCE

Technical Director

Edinburgh University Formula Student (EUPS)

JUL 2026 – PRESENT

- Led a 150-member team across four departments (Mechanical, Electrical, Software, and Business).
- Managed coordination between 18 subteams to work toward achieving the goal of being the first ever in-house manufactured, fully autonomous racecar (ADS) in Formula Student UK.
- Pioneered the team's first bespoke EV from the ground up, in simulation, design, and manufacturing.
- Delivered a functional, track-legal racecar at FSUK26, compliant with autonomous-driving regulations.

Controls Engineering Intern

Horwitz LLC, Building Automation Systems

MAY 2026 – SEP 2026

- Industrial air and fluid handling and corporate HVAC controls. Field and office experience.
- Pioneered the introduction of custom-built BAJA modules within the company. Pioneered LynxSpring Edge controllers running custom BAJA logic engines via BACnet MSTP, including Java trust Auth Cert.
- Trailblazed the company's transition from AutoCAD to Visio. Visual Basic algorithm optimisation for engineering drawings. Decreased company's drawing workflow time by 45%.
- SQL & PowerApps web development. Bespoke web dashboard & mobile app developed to replace Coins ERP. Eliminated the risk of job orders being overlooked, improved job booking efficiency by 20%.
- Sales Engineering: Excel constrained optimisation algorithm to minimise cost for controller I/O config.

Vehicle Dynamics Engineer

Edinburgh University Formula Student (EUPS)

DEC 2024 – JUL 2026

- Designed a suspension geometry optimiser from scratch using Simscape multibodies. Created full-car kinematic model, determined optimal suspension geometry and roll axis. Validated quantitative performance benefits of anti-ackermann and anti-dive and used in-house tyre magic formula model.
- Design, FEA, & topology optimisation for suspension and wheel components in SolidWorks.
- Acted as Suspension Lead at Formula Student UK design presentation, contributed to team's 6th place, topology optimisation and overall design contributed 15 points and mentioned as good design.
- Manufacturing experience, additive manufacturing, CNC, laser cutting, and hydraulic systems.
- Pioneered emergency braking system hydraulic concepts, including actuation and fluid recovery.

Motorboat, Sailboat, and Aircraft Maintenance Technician

Minneapolis Sailing Center (MSC) & Twin Cities Flight Training (TCFT)

JUL 2023 – MAY 2026

- Motorboat & sailboat maintenance at **MSC**, performed hydraulic steering, drivetrain, and electrical maintenance on motorboats, and fibreglass & carbon fibre composites work for our fleet of sailboats.
- Aircraft repairs, teardowns, and component analysis at **TCFT**, performed repairs on hydraulics, tyres, brakes, avionics, and fuselage. Performed teardowns on Cessna airframes and engines.

Sailing Instructor

Minneapolis Sailing Center (MSC)

MAY 2022 – PRESENT

- Head coach for the Optimist Racing Team, managed logistics and coached a team of 8-12 students. Solely responsible for designing all lesson plans and tailoring my coaching for each student.
- Head instructor for the Adult Laser Racing class, taught complex physics and aeronautical concepts, such as wing-tip vortices, airflow separation, and centre of gravity & roll center dynamics.
- 1500+ work hours, 75+ class sessions taught, named best instructor of 2024.



EDUCATION

BEng Honours Mechanical Engineering

The University of Edinburgh

SEP 2024 – MAY 2028

- Grade: First Class (A2). Completed 16 classes with 240 UK credits. On track for first class (~4.0 GPA).

University Classes Taken for High School

University of Minnesota – Twin Cities

SEP 2021 – MAY 2024

- 3.66 Cumulative GPA, 4.0 Technical GPA, 3 Dean's List Honours.
- Completed 17 university classes with 70 US credits. Normal university classes taken full time as secondary school credit. Followed the UMN Mechanical Engineering & Computer Science degree tracks.

Pilot Training

Twin Cities Flight Training (TCFT)

MAY 2022 – JUN 2024

- FAA Licensed Private Pilot. 270 landings, 200 flight hours. Instrument & ILS landings. 30,000+ miles flown.

PROJECTS

Software and Programming Languages

- **MATLAB** Differential equations & matrices calculations for Formula Student.
- **Simulink & Simscape** Simscape multibodies and Simulink for suspension geometry optimiser.
- **SolidWorks** Design, FEA, & topology optimisation for Formula Student.
- **Java** Created Chess, Minesweeper, Tetris, and hashing tables from scratch.
- **Python** Created orbital motion simulations and real-time data processors.
- **HTML & CSS** Used VS-Code to create my personal website: <https://paulhanwang.co.uk>.
- **Niagara 4** BAS controller logic for Lynxspring VAV and JCI JACE. BAJA modules and cert auth.
- **SQL & Power FX** ERP dashboard created in Microsoft PowerApps. Web and mobile app deployment.
- **Visual Basic** Created automated macro workflows for engineering drawings.

Cinematographer & Graphic Designer

- Created several amateur cinematography films and sailboat racing action videos.
- Created bespoke logo, business card, and website design.
- Used Unreal Engine 5 renders for official Edinburgh University Formula Student marketing posters.

Community Outreach Volunteer

- Volunteered at **MSC** as a skipper to give free sailboat rides to members of the public during our Free Sailing Sunday events and local Minneapolis charities during community outreach events.

Academic Outreach Volunteer

- Volunteered for **EUFS** at Blackhall Primary School, giving a presentation on vehicle dynamics and formula student, and helping the class of P6 students to design and 3D print their own racecars.

LICENCES

- **Private Pilot Airplane**  Federal Aviation Administration
- **Motorsport Competition**  Fédération Internationale de l'Automobile
- **CPR-First Aid (Adult and Pediatric)**  American National Red Cross
- **Lake Service Provider Watercraft Operation**  Minnesota Department of Natural Resources
- **OSHA-10**  United States Department of Labor